ROAD POTHOLE DETECTION USING MACHINE LEARNING

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CANDIDATE'S DECLARATION

It is hereby declared that the contents of this thesis are original and any part of it has not been submitted elsewhere for the award of any degree or diploma.

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ABSTRACT

Road potholes are a major cause of traffic accidents, vehicle damage and increased transportation costs, particularly in urban and rural areas with poor road infrastructure. Damaged road surfaces can lead to loss of vehicle control, traffic congestion and serious safety hazards for drivers and pedestrians. Traditional road inspection techniques mainly rely on manual surveys conducted by human inspectors, which are time-consuming, labor-intensively and inefficient for large road networks. These conventional methods also suffer from limited the accuracy and delayed reporting, making it difficult for authorities to identify and repair potholes in a timely manner. To overcome these limitations, this research proposes an automated road pothole detection system using machine learning and computer vision techniques. The road detection systems utilizes object detection models such as YOLOv8, YOLOv11 and Faster R-CNN to detect potholes from road images with high accuracy. These models are trained on labeled datasets and capable of identifying potholes under different lighting and environmental conditions. Experimental results show that machine learning-based detection significantly improves the efficiency of road inspection, reduces manual effort and supports the faster maintenance decisions, ultimately enhancing road safety and infrastructure management.

Keywords: Road Pothole Detection, Machine Learning, Deep Learning, Object Detection, YOLOv8, YOLOv11, Faster R-CNN, Computer Vision, Image Processing, Smart Transportation, Road Safety, Infrastructure Maintenance.

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CHAPTER 1
Introduction

1.1 Overview

Road surface damage, particularly potholes, has become a significant challenge for modern transportation systems, as it directly affects vehicle safety, driving comfort and road durability. Potholes can lead to accidents, increase vehicle repair costs and cause traffic disruptions. Rapid urbanization, heavy vehicle loads and environmental factors such as rainfall and temperature variation contribute to the frequent formation of potholes. Therefore, an efficient and reliable method for identifying road damage is essential for maintaining safe and sustainable transportation infrastructure. Conventional road monitoring approaches are mostly based on manual inspection, which requires substantial time, labor and financial resources. These approaches are often unable to provide continuous and large-scale monitoring of road conditions. To overcome these challenges, this study introduces an intelligent road pothole detection system based on machine learning techniques. Object detection models including YOLOv8, YOLOv11 and Faster R-CNN are employed to analyze road images and automatically detect potholes. The experimental findings indicate that the proposed system achieves effective detection performance, enabling faster maintenance planning and contributing to improved road safety and operational efficiency.

1.2 Problem Statement

Road potholes are a significant threat to traffic safety and vehicle durability, often causing accidents, increased repair costs and traffic disruptions. Existing road inspection methods rely heavily on manual surveys or traditional techniques, which are time-consuming, labor-intensive and ineffective for large-scale and real-time monitoring. Although recent advancements in deep learning and object detection have shown potential for automated

pothole detection, there is still a lack of systematic evaluation of modern detection models such as YOLOv8, YOLOv11 and Faster R-CNN in accurately identifying potholes under varying road and environmental conditions. Differences in detection accuracy, processing speed and computational requirements among these models make it challenging to determine the most suitable approach for practical deployment. Additionally, factors such as lighting variations, road texture complexity and weather conditions further complicate accurate pothole detection. The absence of a reliable and scalable automated system limits timely maintenance actions and increases infrastructure management costs. Therefore, the core problem addressed in this research is to develop an automated road pothole detection system using machine learning and to analyze the effectiveness of different object detection models in improving detection accuracy, operational efficiency and road maintenance decision-making.

1.3 Objectives

The objective of this research is to develop an automated road pothole detection system using machine learning. The study applies and compares YOLOv8, YOLOv11 and Faster R-CNN models to accurately detect potholes from road images and improve road inspection efficiency and safety. The main objectives of this research are as follows:

- To develop an automated road pothole detection system using machine learning and computer vision techniques.
- To collect and prepare a dataset of road images suitable for pothole detection.
- To apply and implement advanced object detection models such as YOLOv8, YOLOv11 and Faster R-CNN for pothole identification.
- To evaluate and compare the performance of different detection models based on accuracy, precision, recall and detection speed.
- To analyze the effectiveness of machine learning-based detection in improving road safety and maintenance efficiency.
- To reduce dependency on manual road inspection methods by providing a faster and more reliable automated solution.

1.4 Social Impact

Social impact refers to the effects that a technology has on people's lives and society as a whole. In the context of this paper, the social impact includes:

Improved Public Safety: Timely detection of road damages such as potholes, cracks, and obstructions helps prevent accidents, especially for pedestrians, cyclists and motorcyclists. Automated deep learning systems enable quick identification and classification of hazards, leading to faster repairs and safer roads.

Enhanced Transportation Efficiency: Damaged roads increase travel time, vehicle wear and tear and fuel consumption. By identifying and addressing road issues early through deep learning, traffic congestion can be minimized and transportation becomes more efficient and reliable.

Cost-Effective Infrastructure Management: Manual inspection is time-consuming and costly. Deep learning automates the damage detection process, reducing human effort and operational costs for municipalities and highway authorities. This allows governments to allocate resources more efficiently.

Accessibility and Equity: By enabling continuous and widespread road monitoring—even in remote or underserved areas—deep learning ensures that maintenance isn't limited to only high-traffic urban roads. This promotes more equitable infrastructure development across different regions.

Environmental Benefits: Efficient road maintenance reduces unnecessary vehicle emissions caused by traffic jams and detours. Also, deep learning models reduce the need for frequent vehicle-based inspection trips, helping to lower carbon footprints.

Data-Driven Urban Planning: Collected road condition data can be used for long-term planning, budget forecasting and prioritizing infrastructure projects. This leads to smarter cities where road infrastructure adapts to real-world usage patterns and degradation trends.

Automatic pothole detection improves road safety and reduces accidents by identifying hazards early. It also helps authorities maintain roads efficiently, saving time and resources.

1.5 Issues Encountered

During the development and implementation of the pothole detection system, several challenges were faced that affected the accuracy and efficiency of the model.

- Poor quality or inconsistent images affecting detection accuracy.
- Limited dataset leading to model underperformance.
- Varying lighting and weather conditions making pothole identification harder.
- High computational requirements for real-time detection.
- Difficulty in differentiating potholes from other road anomalies like shadows or manholes.

Despite these challenges, understanding the issues encountered helps in refining the system and provides guidance for future improvements in pothole detection and road maintenance.

1.6 Organization of the Thesis Report

In the first chapter, the overview, problem statement, social impact, objectives, issues encountered and overall significance of the road pothole detection system are presented to provide a clear understanding of the research motivation and scope. The second chapter contains the literature review, which highlights previous studies conducted in the field of road damage detection, discusses their methodologies, strengths and limitations and establishes the research gap addressed by the present study. The third chapter outlines the research methodology, including data collection techniques, dataset preparation, preprocessing steps, model architecture selection and the overall workflow of the proposed pothole detection system. The fourth chapter describes the system implementation in detail, covering model training, parameter settings, hardware and software requirements and deployment considerations. The fifth chapter presents the experimental results and performance analysis, including a comparative evaluation of YOLOv8, YOLOv11 and Faster R-CNN based on accuracy, precision, recall and detection speed. Finally, the sixth chapter concludes the thesis by summarizing the research findings, discussing key contributions, identifying limitations and

suggesting future research directions to further enhance automated road pothole detection systems.

Overall, this structured organization ensures a logical progression from problem identification to solution development and evaluation. The arrangement of chapters allows readers to clearly understand the research flow, methodology and outcomes. This systematic presentation enhances the readability of the thesis and provides a comprehensive understanding of the proposed road pothole detection system and its practical significance.

1.7 Summary

This chapter presented an overview of road pothole detection and emphasized its importance for ensuring public safety, vehicle durability, and efficient transportation systems. It highlighted the limitations of traditional manual inspection methods and identified the need for an automated, machine learning-based solution. The problem statement discussed the challenges of accurately detecting potholes under varying road, lighting, and environmental conditions and the lack of comprehensive evaluation of modern object detection models. The objectives focused on developing an automated pothole detection system using YOLOv8, YOLOv11 and Faster R-CNN and comparing their performance in terms of accuracy and efficiency. The chapter also explained the social impact of automated detection in improving safety, reducing maintenance costs, and supporting sustainable infrastructure, addressed the key issues encountered during system development, and outlined the organization of the remaining chapters of the thesis.

CHAPTER 2
Literature Review

2.1 Overview

In a literature review, a researcher reviews previous work such as research studies, conference papers, books and articles. This process helps identify what has already been done on the topic, summarize existing knowledge and determine research gaps in earlier studies. Through critical analysis of prior work, limitations can be identified and improved methodologies can be proposed to achieve better research outcomes.

2.2 Background Study

Recent studies have applied deep learning techniques for automated pothole detection to improve accuracy, efficiency and scalability compared to traditional methods. Advanced convolutional neural network–based models enable robust detection under varying road, lighting and environmental conditions, making automated road inspection more reliable.

Fan et al. proposed a multi-scale feature fusion-based semantic segmentation approach to improve pothole detection across varying sizes and complex road conditions [1].

Kang and Kim introduced a dilated convolution-based framework using Faster R-CNN and YOLO models to enhance pothole detection while balancing accuracy and computational efficiency [2].

Lee et al. evaluated pothole detection under low-light conditions using YOLO with feature pyramid networks, achieving improved performance in challenging illumination environments [3].

Jayalakshmi et al. developed a vehicle-mounted real-time pothole monitoring system using YOLOv8, enabling efficient and continuous road inspection [4].

Yurdakul and Taşdemir enhanced YOLOv8 to improve pothole detection and measurement

accuracy while maintaining real-time performance [5].

Zhong et al. combined YOLOv8 with point cloud data to improve pothole detection and depth estimation accuracy [6].

Chetana et al. applied YOLOv11 for intelligent pothole detection, while Ma et al. optimized YOLOv8 for real-time performance, highlighting the suitability of YOLO-based models for practical deployment [7, 8].

Gorro et al. demonstrated that data augmentation techniques significantly improve the robustness of YOLOv8 models under diverse road conditions [9].

Khan et al. proposed a YOLO-based automated road damage detection system suitable for large-scale deployment [10].

Ali et al. improved pothole detection accuracy using Faster R-CNN with multi-scale feature extraction, achieving better localization of small and irregular potholes [11].

Nguyen and Lee introduced a hybrid YOLOv8 and semantic segmentation framework to improve detection reliability in complex road environments [12].

Cheng et al. proposed DeepPotholeNet, a CNN-based architecture designed specifically for pothole detection [13].

Patel and Singh applied transfer learning with YOLOv11 to improve pothole detection on unstructured and rural roads [14].

Park and Kim conducted a comparative study of object detection models, analyzing the trade-off between detection accuracy and processing speed [15].

Overall, these studies demonstrate the effectiveness of modern deep learning models for pothole detection. However, challenges such as lighting variation, occlusion, limited dataset availability, and real-time deployment remain significant research gaps. existing studies demonstrate that deep learning-based approaches, particularly YOLO and Faster R-CNN models, are effective for automated pothole detection under diverse road and environmental conditions. While YOLO-based models are well suited for real-time detection due to their high processing speed, Faster R-CNN offers improved localization accuracy at the cost of higher computational requirements. However, variations in datasets, environmental conditions and evaluation metrics make direct comparison challenging. These limitations highlight the need for a systematic evaluation of modern object detection models to identify an optimal balance between accuracy and efficiency for practical road pothole detection systems.

Table 2.1: Comparison of Related Works on Pothole Detection

Author (Year)	**Approach / Focus**	**Model**	**Key Findings and Limitations**
Fan et al. (2021) [1]	Semantic segmentation using multi-scale feature fusion	Segmentation CNN	Dataset: Pothole-600. Improved mIoU over baseline models but not suitable for direct object detection.
Kang and Kim (2021) [2]	Dilated convolution for real-time pothole detection	Faster R-CNN, YOLO-based	Dataset: Custom road dataset. Precision $\sim$91.9%, but performance decreases in harsh lighting and weather conditions.
Lee et al. (2023) [3]	Low-light pothole detection using FPN and augmentation	YOLO-based model	Dataset: Low-light pothole dataset. mAP@50 $\sim$0.88 but limited real-time deployment evaluation.
Jayalakshmi et al. (2025) [4]	Vehicle-based monitoring using GPS and cameras	YOLOv8	Dataset: Mobile road dataset. Improved efficiency over YOLOv2 but lacks detailed quantitative comparison.
Yurdakul and Taşdemir (2025) [5]	Enhanced YOLOv8 architecture for detection and measurement	Improved YOLOv8	Dataset: Road pothole dataset. High precision and real-time performance but requires validation on larger datasets.

Author (Year)	**Approach / Focus**	**Model**	**Key Findings and Limitations**
Zhong et al. (2025) [6]	YOLOv8 combined with point cloud fusion	YOLOv8 + Sensor Fusion	Dataset: RGB + point cloud dataset. Improved quantification but requires specialized sensors and hardware.
Chetana et al. (2025) [7]	Intelligent pothole detection pipeline	YOLOv11	Dataset: Road damage dataset. Improved detection accuracy but limited real-world testing.
Ma et al. (2024) [8]	Optimized YOLOv8 feature layers for real-time detection	Modified YOLOv8	Dataset: Custom dataset. Higher mAP than baseline YOLOv8 but still affected by occlusion and shadows.
Gorro et al. (2024) [9]	Image augmentation for robust detection	YOLOv8	Dataset: Augmented pothole dataset. Improved robustness but dataset size remains limited.
Khan et al. (2024) [10]	YOLO-based automated road damage detection	YOLO variants	Dataset: Public road damage datasets. High detection performance but limited cross-region generalization.
Ali et al. (2024) [11]	Multi-scale feature extraction for pothole detection	Faster R-CNN	Dataset: Road pothole dataset. Improved precision and recall but computationally expensive for real-time use.

Author (Year)	**Approach / Focus**	**Model**	**Key Findings and Limitations**
Nguyen and Lee (2025) [12]	Hybrid YOLOv8 and semantic segmentation framework	YOLOv8 + Segmentation	Dataset: Mixed road damage dataset. Better detection in complex scenes but increased training complexity.
Cheng et al. (2025) [13]	DeepPotholeNet CNN architecture	Custom CNN	Dataset: Road pothole dataset. High accuracy but lacks real-time performance evaluation.
Patel and Singh (2024) [14]	Transfer learning for unstructured roads	YOLOv11	Dataset: Unstructured road dataset. Robust detection but requires more benchmark comparisons.
Park and Kim (2025) [15]	Comparative evaluation of object detection models	Multiple detectors	Dataset: Road damage benchmark dataset. YOLO provides best speed–accuracy trade-off but performance varies by dataset.

2.2.1 Discussion of Related Works

The reviewed literature demonstrates significant advancements in automated pothole detection using deep learning techniques. Many studies have focused on enhancing the feature extraction capabilities and real-time performance of object detection models. For instance, Fan et al. [1] proposed a multi-scale feature fusion strategy for semantic segmentation, which improved the understanding of pothole regions. Similarly, Kang and Kim [2] integrated dilated convolution with Faster R-CNN and YOLO-based models, achieving better feature extraction while maintaining real-time detection capabilities. Several studies emphasized improving detection under challenging conditions. Lee et al. [3] evaluated YOLO models with feature pyramid networks in low-light environments and showed improved nighttime detection performance. Vehicle-mounted systems have also been explored to enable on-the-go road monitoring. Jayalakshmi et al. [4] combined YOLOv8 with real-time inspection, whereas Yurdakul and Taşdemir [5] enhanced the YOLOv8 architecture for more accurate pothole detection and measurement. Additionally, Zhong et al. [6] integrated YOLOv8 with point cloud fusion, enabling better pothole quantification.

The evolution of YOLO-based models is evident across multiple studies. Chetana et al. [7] utilized YOLOv11 for intelligent pothole identification, and Ma et al. [8] proposed an improved YOLOv8 model for faster and more accurate real-time detection. Gorro et al. [9] highlighted the importance of data augmentation to improve the robustness of YOLOv8, while Khan et al. [10] presented a comprehensive automated road damage detection system using various YOLO variants. In parallel, Faster R-CNN approaches remain relevant. Ali et al. [11] applied Faster R-CNN with multi-scale feature extraction, improving detection accuracy. Hybrid approaches combining object detection with semantic segmentation have also been investigated, as demonstrated by Nguyen and Lee [12] for complex road environments. Novel architectures, such as DeepPotholeNet introduced by Cheng et al. [13], show the potential of CNN-based networks specifically tailored for pothole detection. YOLOv11 combined with transfer learning, as used by Patel and Singh [14], provides effective detection on unstructured roads. Finally, Park and Kim [15] compared multiple object detection models, emphasizing the trade-off between detection speed and accuracy.

Overall, these studies collectively confirm the effectiveness of modern deep learning models, particularly YOLO and Faster R-CNN variants, for pothole detection. However, several

challenges remain, including lighting variations, occlusions, limited annotated datasets, and real-time deployment constraints, which represent ongoing research gaps in this field.

2.3 Summary

Road damage detection research generally follows three major approaches: classification-based, detection-based and technology-driven systems. Classification-based approaches identify the type of road damage from images but do not provide precise localization. Detection-based methods, such as Faster R-CNN and YOLO localize potholes using bounding boxes and confidence scores and are commonly enhanced through transfer learning and data augmentation techniques. Technology-driven approaches integrate deep learning models with real-world systems such as smartphone or dashcam-based data collection, OpenCV-based preprocessing and deployment on mobile or embedded platforms, enabling fast, low-cost and real-time pothole detection solutions.

CHAPTER 3
Methodology

3.1 Overview

This study detects potholes using deep learning object detection. First, road images are collected and annotated with pothole bounding boxes, then split into training, validation and test sets. The images are resized and augmented (e.g., flip, brightness, blur) to improve robustness. Next, YOLOv8, YOLOv11 and an R-CNN-based model are trained using transfer learning. Finally, the models are evaluated using precision, recall and mAP and the best model is selected for pothole detection on unseen images.

3.2 Data Collection and Preprocessing

Data collection and preprocessing are crucial steps in developing an effective road pothole detection system. In this study, a dataset of road images was collected from publicly available sources and real-world road scenes captured using cameras under different environmental conditions, including varying lighting, weather and road textures. The dataset includes images containing potholes of different sizes, shapes, and severity levels, as well as non-pothole road surfaces to ensure balanced learning. All images were manually annotated using bounding boxes to label pothole regions, making the dataset suitable for supervised object detection models such as YOLOv8, YOLOv11 and Faster R-CNN.

Data collection and preprocessing are closely connected steps in the pothole detection system. Road surface images are first collected using cameras or mobile devices under different road and lighting conditions to capture diverse pothole appearances, including variations in size, shape, and depth. These collected images are then preprocessed by resizing them to a uniform size, normalizing pixel values, and annotating potholes with appropriate labels to create reliable ground truth data. Data augmentation techniques such as flipping and rotation may also be

applied to increase dataset diversity. Finally, the prepared dataset is divided into training, validation and testing sets, ensuring the data is suitable for effective model training, model tuning and accurate performance evaluation. The Figure 3.1 shows our data preprocessing workflow for the pothole detection dataset.

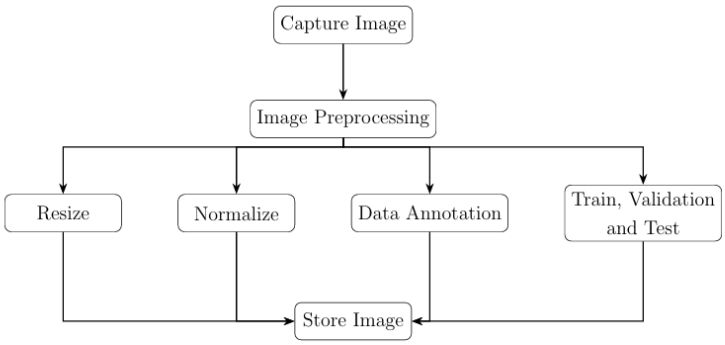


Figure 3.1: Flow Chart of Data Preprocessing

First, we capture road images and then perform image preprocessing. In this step, we resize the images to a fixed input size and normalize pixel values to maintain consistency. Next, we complete data annotation by labeling potholes and then split the dataset into training, validation, and test sets. Finally, we store the processed and labeled images for model training and evaluation.

3.2.1 Data Sources

We used road images with potholes and without potholes. The dataset was collected from publicly available sources including Kaggle, Roboflow and GitHub, along with real-world road images captured using mobile and dashboard cameras. Additional images were sourced from online repositories and manually annotated using bounding box labeling tools.

The data used for pothole detection consists of road surface images collected from multiple sources. Images are captured using cameras or mobile devices mounted on vehicles, ensuring coverage of different road types and environmental conditions. In addition, publicly available road and pothole image datasets may be incorporated to increase data diversity. These combined data sources help improve the robustness and generalization ability of the pothole detection model.

The images were collected using cameras or mobile devices and include variations in road texture, pothole size, shape, lighting, and shadows. This raw dataset serves as the primary

source for pothole detection and provides realistic samples required for effective training and evaluation of the proposed model. This Figure 3.2 show the raw dataset consists of road surface images containing visible potholes captured under real-world conditions.

Figure 3.2: Raw collected Dataset

1. Publicly Available Online Datasets

Road pothole images were obtained from open-source datasets available on platforms such as Kaggle, Roboflow Universe, and GitHub, which provide annotated images of road damages collected under different environmental conditions.

2. Real-World Image Collection

Additional road images were captured using mobile phone and dashboard cameras mounted on vehicles. These images were collected from various road environments, including urban and semi-urban areas, under different lighting and weather conditions to improve dataset realism.

3. Web-Based Image Sources

Supplementary pothole images were gathered from publicly accessible web sources such as Google Images to increase variation in pothole shapes, sizes, and road textures.

4. Manual Annotation

All collected images were manually annotated using labeling tools such as LabelImg and

Roboflow Annotate, where bounding boxes were drawn around pothole regions to create a supervised learning dataset.

3.2.2 Methodology

The methodology shown in Figure 3.3 for road pothole detection follows a systematic pipeline consisting of image collection, preprocessing, model training, testing and final detection output. Initially, road images are collected from publicly available datasets and real-world sources using mobile phones and dashboard cameras mounted on vehicles. These images are captured under diverse environmental conditions, including different lighting, weather and road surface variations, to ensure dataset diversity and robustness.

After preprocessing, the prepared dataset is used to train object detection models including YOLOv8, YOLOv11 and Faster R-CNN. These models learn to detect and localize potholes in road images. The trained models are evaluated using a separate testing dataset to measure detection accuracy, precision, recall and inference speed. Finally, the system generates pothole detection outputs in the form of bounding boxes, enabling effective comparison of model performance and supporting automated road inspection and maintenance planning.

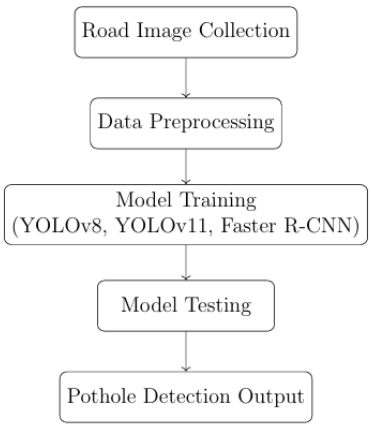


Figure 3.3: Methodology

Algorithm Steps:

1. Collect road images from public datasets and real-world sources.
2. Annotate pothole regions in images using bounding boxes.
3. Preprocess images by resizing and normalizing pixel values.
4. Apply data augmentation to enhance dataset diversity.
5. Split the dataset into training, validation and testing sets.

6. Train YOLOv8, YOLOv11 and Faster R-CNN models using the training dataset.
7. Test the trained models on unseen images.
8. Evaluate model performance using accuracy, precision, recall and detection speed.
9. Generate pothole detection output with bounding box localization.

3.2.3 Data Collection Techniques

Data for this study was collected using multiple techniques to ensure diversity and reliability. Publicly available datasets were obtained from online platforms such as Kaggle and Roboflow, which provide road images captured under various conditions. In addition, real-world images were collected using mobile phones and dashboard cameras mounted on vehicles while driving on different types of roads. Images were captured under varying environmental conditions, including daytime, nighttime, rainy weather and different road surfaces, to improve model robustness. All collected images were manually annotated using labeling tools to generate accurate bounding boxes, making the dataset suitable for supervised machine learning models.

3.2.4 Road Pothole Detection Using YOLO

This study uses YOLOv8 and YOLOv11 for automated road pothole detection. Both models are trained on annotated and preprocessed road images to detect potholes in real time. They efficiently identify potholes of different sizes and shapes under varying road and environmental conditions, with performance evaluated using accuracy, precision, recall and detection speed. YOLOv8 is employed for real-time pothole detection on annotated and preprocessed road images. It can efficiently detect potholes of various sizes and shapes under different lighting and weather conditions. The model's performance is evaluated using accuracy, precision, recall and detection speed.YOLOv8 is a state-of-the-art, single-stage object detection model designed for fast and accurate detection. In the YOLOv8-based pipeline, the input image is first processed by the CSPDarknet backbone, which extracts rich spatial and semantic features from the image. These features are then passed to the neck, which uses Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) to fuse multi-scale features, enabling effective detection of objects of different sizes.

The detection head in YOLOv8 predicts bounding boxes and class labels directly from the fused features without using region proposal networks. Finally, post-processing steps such as non-maximum suppression are applied to generate the final detections, making YOLOv8

suitable for real-time applications like road pothole detection due to its high speed and accuracy. This Figure 3.4 the architecture diagram represents the object detection pipeline used in the proposed pothole detection system.

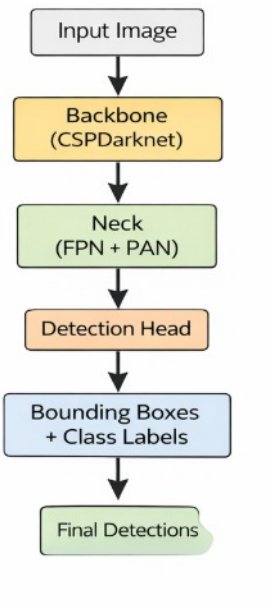


Figure 3.4: Architecture used in YOLOv8

The process starts with an input image, which is passed through the backbone (CSPDarknet) to extract important visual features. These features are then processed by the neck (FPN + PAN) to combine multi-scale information. The detection head uses these fused features to predict bounding boxes and class labels, which are finally refined to produce the final detections of potholes.

YOLOv11 is also applied for automated pothole detection. It improves detection robustness in complex road scenarios and ensures fast and accurate identification of potholes. Its effectiveness is measured through the same performance metrics, enabling comparison with YOLOv8. YOLOv11 is an advanced real-time object detection model designed to achieve higher accuracy and efficiency compared to earlier YOLO versions. In a YOLOv11-based system, the input image is first processed by the CSPDarknet backbone, which extracts deep and meaningful features from the image. The backbone incorporates improved modules such as ELAN and blocks to enhance feature representation while maintaining computational efficiency. The extracted features are then passed to the neck, which uses Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) to fuse multi-scale features, allowing

effective detection of potholes of varying sizes. The dense prediction head predicts bounding boxes, class labels and optionally segmentation masks directly from these features. Finally, post-processing techniques refine the predictions to produce accurate final detections, making YOLOv11 well-suited for real-time and robust pothole detection applications. This Figure 3.5 diagram illustrates the YOLOv11 detection pipeline.

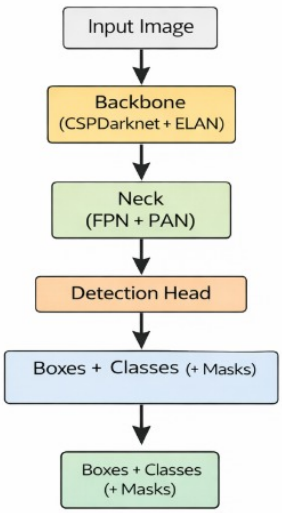


Figure 3.5: Architecture used in YOLOv11

The process begins with an input image, which is passed through the backbone (CSPDarknet with ELAN modules) to extract rich and discriminative features. These features are then processed by the neck (FPN + PAN) to fuse multi-scale information, improving detection of objects of different sizes. The detection head uses the fused features to predict bounding boxes and class labels, and optionally segmentation masks. Finally, the refined outputs represent the final detections, indicating the location and class of potholes in the image.

3.2.5 Road Pothole Detection Using Faster R-CNN

Faster R-CNN is used for automated pothole detection by extracting features from road images and proposing candidate regions for potholes. It provides accurate detection of potholes of different sizes and shapes, even under varying road and environmental conditions. Performance is evaluated using metrics such as accuracy, precision, recall, and detection speed, allowing comparison with YOLO models. The performance of the Faster R-CNN model is evaluated using metrics such as accuracy, precision, recall and detection speed, enabling a comprehensive comparison with YOLO-based models. Although Faster R-CNN typically requires higher

computational resources, its two-stage detection approach ensures improved localization accuracy and reduced false detections. This Figure 3.6 diagram represents the Faster R-CNN object detection pipeline.

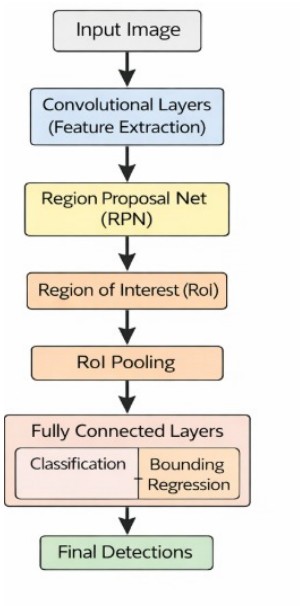


Figure 3.6: Architecture used in Faster R-CNN

The process starts with an input image, which is passed through convolutional layers to extract feature maps. A Region Proposal Network (RPN) then generates candidate object regions, known as Regions of Interest (RoIs). These regions are refined using RoI pooling to obtain fixed size feature representations. The features are then passed through fully connected layers for classification and bounding box regression, resulting in the final detections of objects such as potholes.

3.2.6 Annotation Process

The annotation process involves labeling potholes in collected road images using bounding boxes. Each pothole is manually marked to create a ground truth dataset, which is essential for supervised learning. Tools such as Roboflow annotation tool are used to ensure precise and consistent labeling, making the dataset suitable for training object detection models like YOLOv8, YOLOv11, and Faster R-CNN.

The annotation process involves labeling pothole regions in road surface images to create ground truth data for supervised learning. Each image is manually annotated using an annotation tool by drawing bounding boxes or polygon masks around visible potholes. Care

is taken to accurately outline the pothole boundaries while avoiding irrelevant background regions. The annotated images are reviewed to ensure consistency and correctness. These annotations are then saved in a standard format compatible with the detection model, enabling effective training, validation, and evaluation of the pothole detection system. This Figure 3.7 shows the annotation process of the pothole dataset.

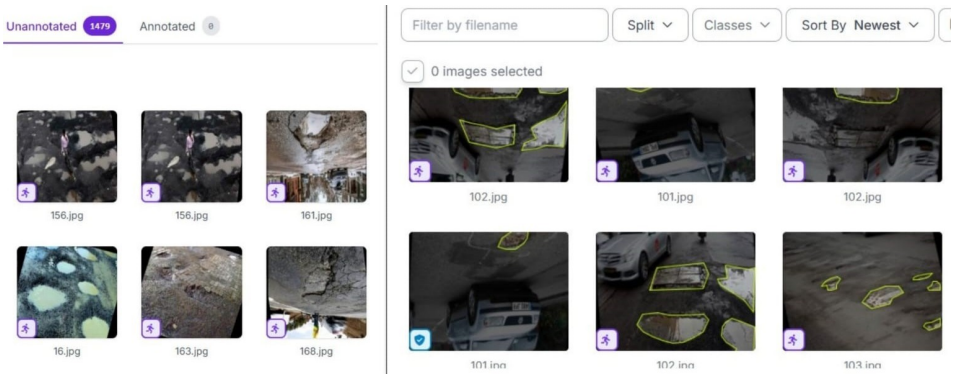


Figure 3.7: Annotated and Unannotated dataset of road potholes

On the left side, unannotated road images are shown, where potholes are visible but not yet labeled. On the right side, the same type of road images are displayed after annotation, with potholes clearly marked using yellow outlines. These annotations highlight the exact location and shape of potholes, converting raw images into labeled data suitable for training and evaluating object detection models.

3.2.7 Dataset Description

In this study, a custom dataset was created consisting of approximately 3,878 images of road potholes or damage roads of different areas.The dataset consists of road images collected from public sources (e.g., Kaggle, Roboflow) and real-world captures using mobile phones or vehicle-mounted cameras. It includes potholes of varying sizes, shapes, and severity, along with non-pothole road surfaces for balanced learning. All images are annotated with bounding boxes, preprocessed with resizing, normalization, and augmentation and divided into training, validation and testing sets for model evaluation.

3.3 Pre-Trained Models for Comparative Analysis

Pre-trained models such as YOLOv8, YOLOv11 and Faster R-CNN are used to compare performance in road pothole detection. These models are trained on large datasets beforehand and fine-tuned on the collected pothole dataset. Using pre-trained models helps reduce training time, improves detection accuracy, and allows a fair comparison of precision, recall and detection speed under different road and environmental conditions. The Table 3.2 compares the performance of three object detection models—YOLOv8, YOLOv11 and Faster R-CNN—using metrics like precision, recall, mAP50, and mAP50–95.

Table 3.2: Performance Comparison of Detection Models

Model Name	**Precision (%)**	**Recall (%)**	**mAP50 (%)**	**mAP50–95 (%)**
YOLOv8	93.7	92.3	97.5	82.1
YOLOv11	93.1	92.9	97.3	81.1
Faster R-CNN	46.4	46.6	62.1	35.3

Precision indicates how many detected objects are correct, while recall shows how many actual objects are detected. mAP50 measures accuracy at an IoU of 0.5, and mAP50–95 provides a stricter evaluation across multiple IoU thresholds. These metrics together give a clear view of each model's detection capabilities.

From the results, YOLOv8 performs the best, achieving the highest precision (93.7%) and recall (92.3%), with strong mAP50 (97.5%) and mAP50–95 (82.1%). YOLOv11 is close behind with slightly lower values but still reliable. Faster R-CNN shows much weaker performance, with around 46% precision and recall and significantly lower mAP scores. Overall, YOLOv8 is the most accurate and efficient model for object detection, followed by YOLOv11, while Faster R-CNN lags considerably.

3.4 Summary

The methodology of road pothole detection system follows a systematic pipeline. First, road images are collected from public datasets, real-world captures, and web sources, covering diverse lighting, weather, and road conditions. All images are manually annotated with bounding boxes and preprocessed through resizing, normalization, and augmentation to improve model robustness. Next, pre-trained object detection models—YOLOv8, YOLOv11

and Faster R-CNN—are fine-tuned on the prepared dataset. The models are then evaluated using accuracy, precision, recall, mAP and detection speed and the best-performing model is selected for automated pothole detection on unseen road images.

CHAPTER 4
Implementation

4.1 Overview

the main goal of our thesis is to put the suggested strategy into practice. Using these methods, we can Detect the potholes of routes from an image. However, implementing this strategy is a complex and challenging process. For the implementation of the method, we used Pacific tools and technologies.

4.2 Technologies Used

Implement and evaluate the road pothole system, a combination of hardware and software technologies was used. These tools support various stages of the project including data collection, data preprossessing, Data annotation, model training end result analysis.

4.3 Dataset Preparation

A smartphone was used as the primary hardware device to capture images of real-world road potholes. This approach is practical, cost-effective, and aligned with real-world deployment scenarios, where pothole detection systems are often integrated into mobile or vehicle-mounted devices. Capturing images from different angles and distances also helped incorporate variations in pothole size, shape, and severity.

After image acquisition, the dataset was prepared using modern computer vision tools. All collected images were uploaded to the Roboflow platform for annotation and pre-processing. Bounding boxes were manually drawn around pothole regions to generate accurate ground truth labels suitable for supervised object detection. Roboflow was further used for image

resizing, normalization, and data augmentation techniques such as flipping, rotation, and brightness adjustment. These pre-processing steps increased dataset diversity and reduced the risk of model overfitting.

For model training and experimentation, Kaggle was used as the cloud-based development environment. Kaggle provides free access to GPU resources, making it suitable for training deep learning models such as YOLOv8, YOLOv11, and Faster R-CNN. The platform also facilitated dataset management, model prototyping, evaluation, and performance comparison within a structured and reproducible workflow.

Overall, the dataset preparation process ensured that the collected road images were properly annotated, preprocessed, and organized for effective training, validation, and testing of the pothole detection models.

4.4 Model Selection

The popular CNN architectures listed below have been selected for assessment:

YOLOv8: YOLOv8 Is there one stage object detection model developed ultralytics. It performs object localization and classification in a single forward making it highly suitable for real time applications such as road pothole detection. YOLOv8 Uses an anchor-free detection mechanism, which simplifies bounding box prediction and improves detection accuracy. The architecture consists of a backbone for future extraction, And a detention head predicting bounding boxes and class probabilities. Due to its high inference speed and competitive accuracy, YOLOv8 is widely used in real-time computer vision system.

YOLOv11: YOLOv11 is a newer generation of YOLO model designed to further improve accuracy, efficiency, and scalability compared to previous version. YOLOv11 improves upon models by optimizing feature extraction and refining the detection head, resulting in better performance under challenging conditions such as poor lighting uneven road textures. Despite higher accuracy, YOLOv11 maintenance efficient inference speed, making it a strong candidate for both real-time and near real-time applications.

Faster R-CNN: Faster R-CNN (Region-based Convolutional Neural Network) Is it two stage object detection model. In the first stage, a Region Proposal Network (RPN) generates candidate object regions. In the second stage, these regions are classified and refined to produce final bounding boxes. Faster R-CNN Is known for its high detection accuracy especially in complex scenes However due to its two-stage architecture, it has slower inference speed compared to YOLO models These make it more suitable for offline analysis rather than real time detection. In road pothole detection, Faster R-CNN often achieve higher precision but at the cost of increased computational complexity.

4.5 Implementation Steps

The implementation process involved adapting pre trained convolutional neural network architectures for road pothole detection task. The key steps were as follows:

1. **Dataset annotation:** All the image data sets were uploaded in rubber flow, from there the images were labeled with classes, preprocessed and augmented.

2. **Load pre trained model:** The backbone of each model was loaded In the Kaggal environment.

3. **Freeze initial layers:** The early convolutional layers were frozen to retain previous learned features.

4. **Fine tuning:** After initial training of the Pre train model, The last few convolutional layers were unfrozen and fine tuned with a lower learning rate

4.6 Training Configuration

The Table 4.3 defines the key hyperparameters and strategies used to optimize The Deep Learning models effectively. The choice of the optimizer, loss function, learning rate, batch size and number of epochs directly impact convergence speed and final model accuracy. Stochastic Gradient Descent (SGD) Optimizer is selected Because optimized for speed and accuracy balance, performs well for small object detections like potholes.

Table 4.3: Training Configuration

Parameter	**Value**
Optimizer	Stochastic Gradient Descent
Loss Function	Box Regression Loss & RPN Regression Loss
Learning Rate	0.001
Batch size	32
Epochs	70
Early stopping	15
Validation split	20%

4.7 Model Evaluation

After training, each model was evaluated on the unseen test set using the following matrix. This section present the evaluation matrix used to this is the performance of the deep learning model in road pothole detection. These include Accuracy, Recall, F1-score and the Confusion Matrix. These metrics provide a comprehensive understanding of the models classification capabilities and their effectiveness in detecting various type of road potholes.

- **Accuracy:** Accuracy refers to the ratio of correctly predicted instances to the total number of prediction.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4.1}$$

  where,

  TP = True Positive, TN = True Negative

  FP = False Positive, FN = False Negative

  Although accuracy is common use metrics. It may be misleading in imbalanced dataset, wire one class significantly outnumbers others.

- **Precision:** Precision measures how many of the previously predicted cases are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4.2}$$

A high precision indic a low false positive rate. It is useful when the cost of false positive is high.

- **Recall:** Recall calculates how many actual positive instances were correctly predicted.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4.3}$$

High recall is crucial when missing a positive case Significant consequences.

- **F1-score:** The F1-score is the harmonic mean of precision and recall it provides a balance between the two, especially when the data set is imbalanced.

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4.4}$$

A higher F1-score indicates better model performance in terms of both precision and recall.

- **Confusion Matrix:** The confession matrix table 4.4 is a performance measurement tool for classification problems It is especially useful for visual multi-class classification results.

Table 4.4: Confusion Matrix (Binary Classification)

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive	False Negative
Actual Negative	False Positive	True Negative

In multi class classification, the confusion matrix extends to an ***n x n*** grid, where *n* is the number of classes. Each cell *(i,j)* represent the number of times instance of class *i* were predicted as class *j*

4.8 Summary

Each model was successfully trained and evaluated using the same pipeline. The dataset was carefully preprocessed and augmented to improve performance. Transfer allowed efficient use of pre-trained models, with YOLOv8 achieving the best result.

CHAPTER 5
Result and Analysis

5.1 Overview

We have tested deep learning models on our dataset. the tested models are YOLOv8, YOLOv11 and Faster R-CNN. The preformation of all these models were used to measure the accuracy of different models. Among them YOLOv8 provided the best accuracy.

5.2 Experiment Result

5.2.1 YOLOv8

We have 97.7 % accuracy using YOLOv8 model. in Figure 5.8, illustrates the training and validation losses curve over 70 epochs. The training loss remains lower than the validation loss throughout, while both accuracy curves a consistent upward trend indicating effective learning and good generalization capability of the model. Figure 5.9 presents the F1 confidence curve, precision recall curve, precision confidence curve, recall confidence curve. Figure 5.10 present the confusion matrix of the model evaluated on 3867 samples. The test and validation data contains a total image of 1242, equally distributed across the class ”road_pothole”.

Training and Validation performance for YOLOv8

The Figure 5.8 shows the P, R losses over 70 epochs. The Figure 5.9 shows the F1-Confidence curve, Precision-Recall curve, Precision-Confidence curve, Recall-Confidence curve of the YOLOv8 model.

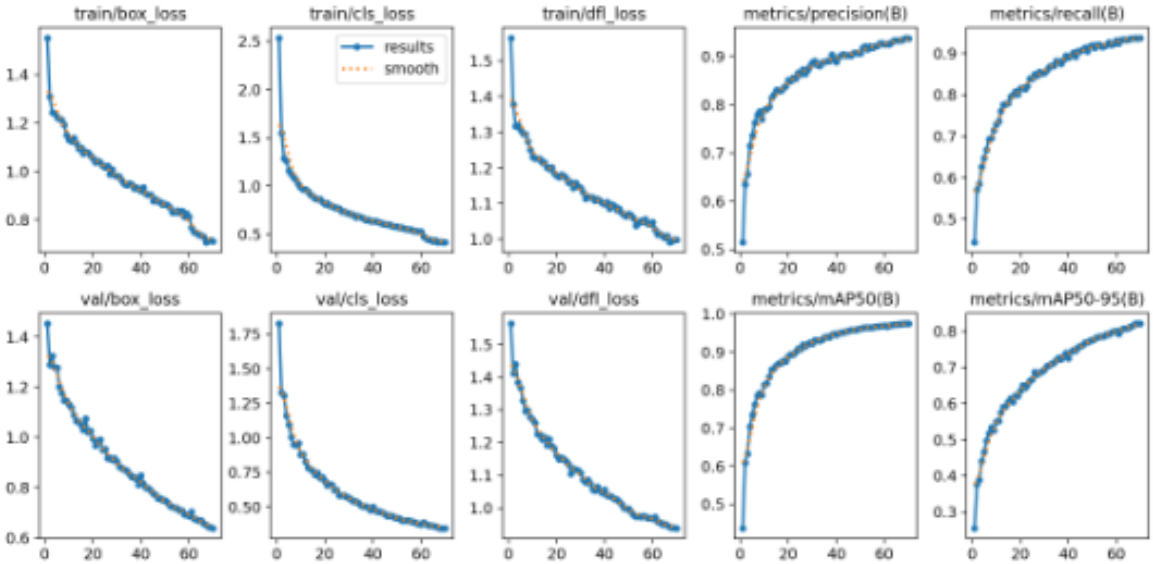


Figure 5.8: Training, Validation Curve (P, R, Losses)

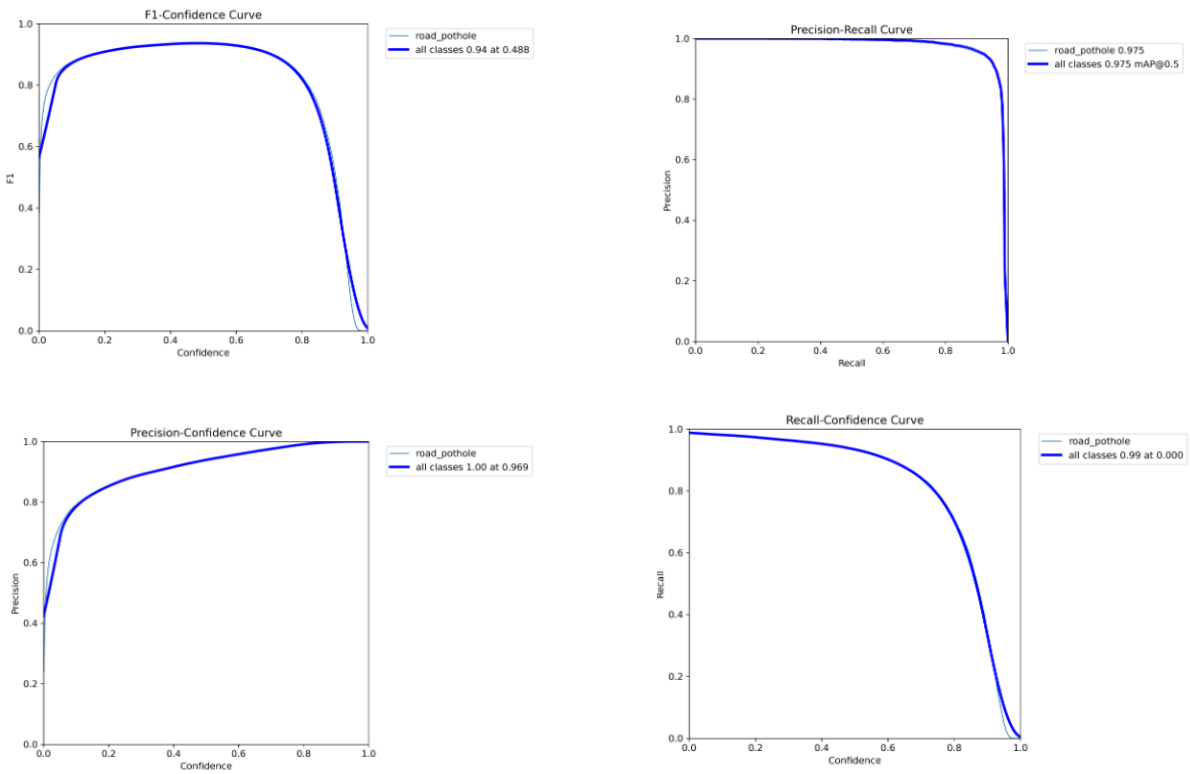


Figure 5.9: F1-Confidence, Precision-Recall, Precision-Confidence, Recall-Confidence curve

Classification Report for YOLOv8

In the Table 5.5 we can see that YOLOv8 achieved an overall accuracy of 97.3%, Precision 93.7%, Recall 93.9%, mAP_{50} 97.5%, mAP_{50-95} 82.2%. The F1 score of the model was also

excellent of 93.7%

Table 5.5: Evaluation Report of YOLOv8

Parameter	**Value**
Precision	93.7%
Recall	93.9%
mAP_{50}	97.5%
mAP_{50-95}	82.2%
F1-score	93.7
Macro Average	93.7%
Weighted Average	93.7%
Accuracy	97.7%

Confusion Matrix of YOLOv8

The confusion matrix Figure 5.10 revealed near perfect classification for road porthole with minor misclassification between road pothole and background. The primary challenge lies in distinguishing wrote porthole and background.

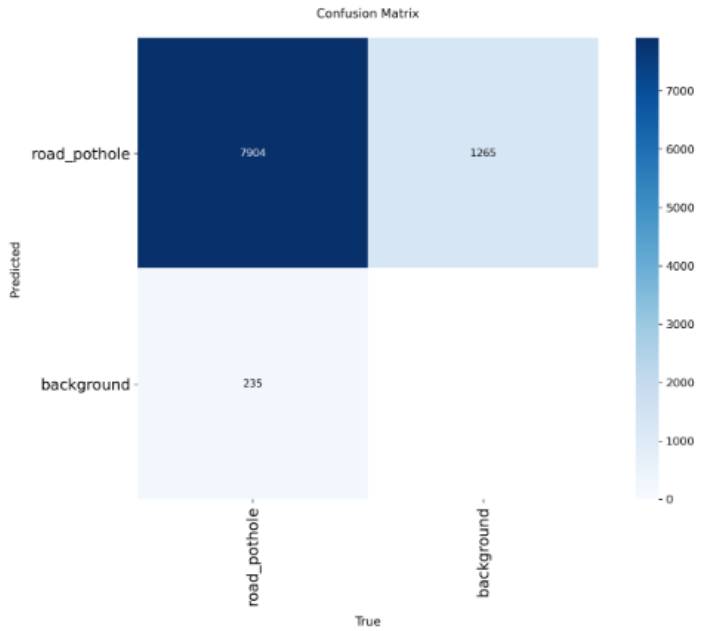


Figure 5.10: Confusion Matrix of Yolov811

5.2.2 YOLOv11

We have 97.3 % accuracy using YOLOv11 model. in Figure 5.11, illustrates the training and validation losses curve over 70 epochs. The training loss remains lower than the validation loss throughout, while both accuracy curves a consistent upward trend indicating effective learning and good generalization capability of the model. Figure 5.12 presents the F1 confidence curve, precision recall curve, precision confidence curve, recall confidence curve. Figure 5.13 present the confusion matrix of the model evaluated on 3867 samples. The test and validation data contains a total image of 1242, equally distributed across the class ”road_pothole”.

Training and Validation performance for YOLOv11

The Figure 5.11 shows the P, R losses over 70 epochs. The Figure 5.12 shows the F1-Confidence curve, Precision-Recall curve, Precision-Confidence curve, Recall-Confidence curve of the YOLOv11 model.

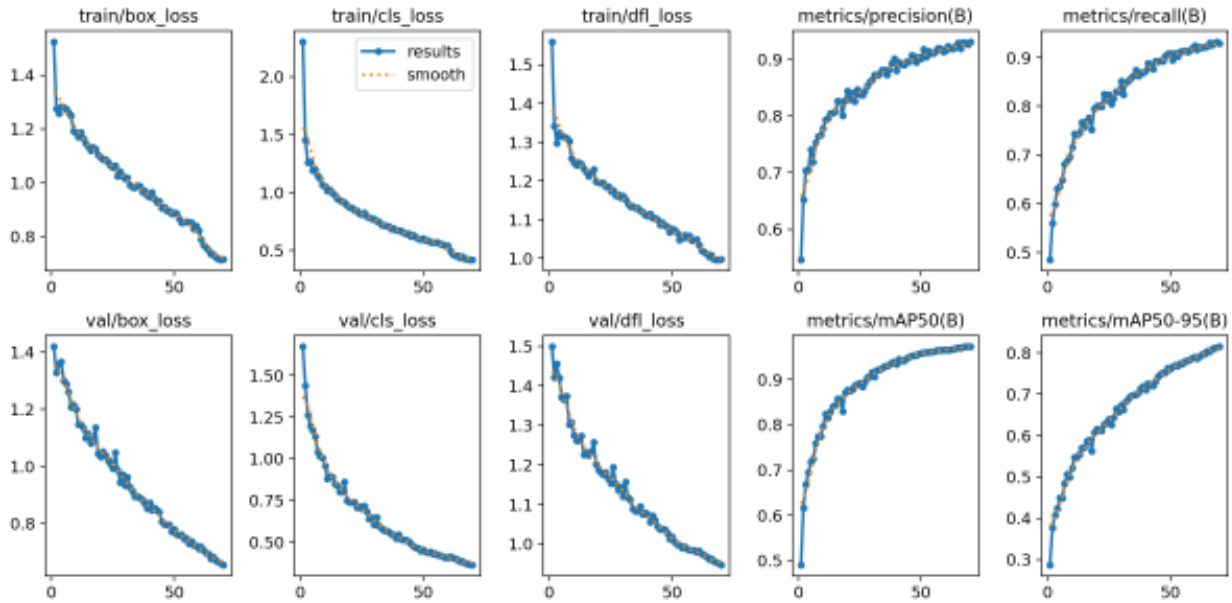


Figure 5.11: Training, Validation Curve (P, R, Losses)

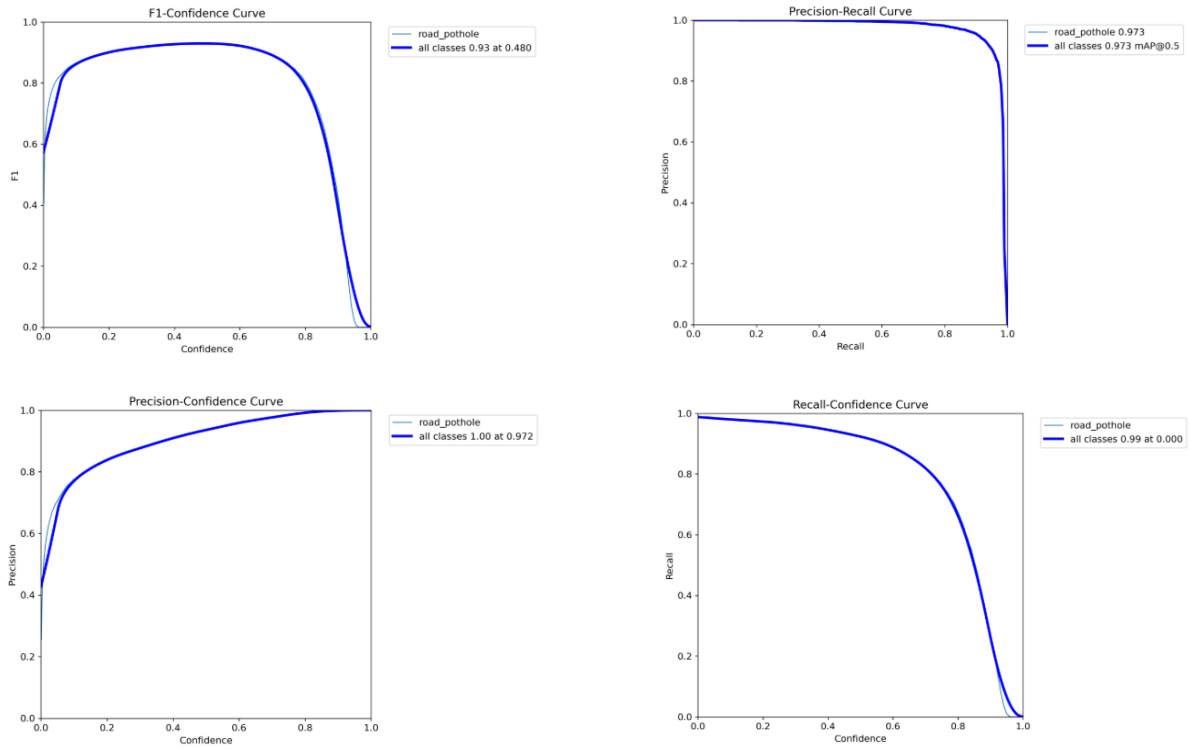


Figure 5.12: F1-Confidence, Precision-Recall, Precision-Confidence, Recall-Confidence curve

Classification Report for YOLOv11

In the Table 5.6 we can see that YOLOv11 achieved an overall accuracy of 97.3%, Precision 93.1%, Recall 92.9%, mAP_{50} 97.3%, mAP_{50-95} 81.6%. The F1 score of the model was also excellent of 93.0%

Table 5.6: Evaluation Report of YOLOv11

Parameter	**Value**
Precision	93.1%
Recall	92.9%
mAP_{50}	97.3%
mAP_{50-95}	81.6%
F1-score	93.0
Macro Average	93.1%
Weighted Average	93.1%
Accuracy	97.3%

Confusion Matrix of YOLOv11

The confusion matrix Figure 5.13 revealed near perfect classification for road porthole with minor misclassification between road pothole and background. The primary challenge lies in distinguishing wrote porthole and background.

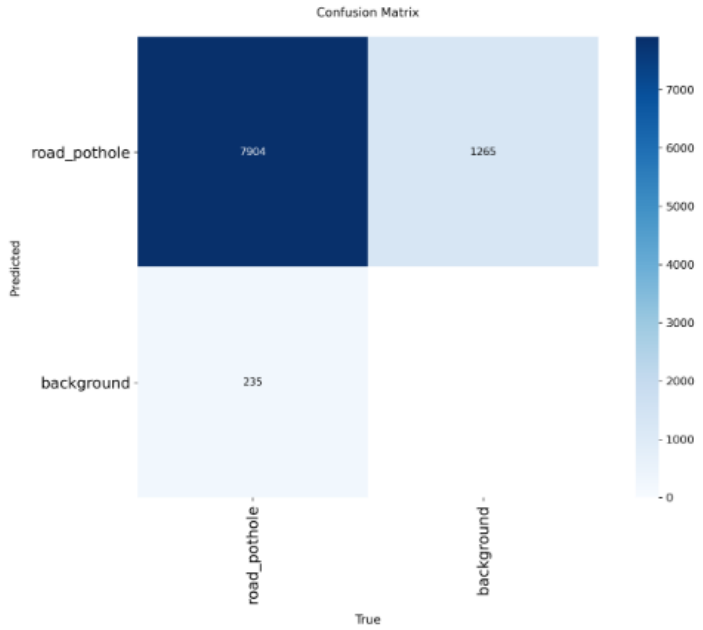


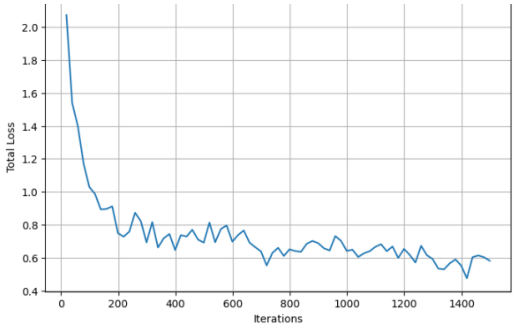
Figure 5.13: Confusion Matrix of YOLOv11

5.2.3 Faster R-CNN

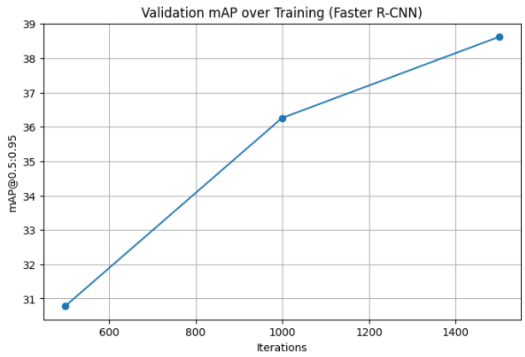
We have 62.1% of global accuracy using Faster R-CNN model. in Figure 5.14, illustrates the training loss and validation mAP curve over 1500 iteration. The model shows that it can detect bigger pothole successfully although struggles to detect small potholes. The table shows the evaluation score of Precision, Recall, mAP_{50}, mAP_{50-95} of the model.

Training and validation performance for Faster R-CNN

The Figure 5.14 shows the training losses and validation curve over 1500 iterations, training loss decreased steadily with minimal gap and the mAP_{50-95}, which is the average precision across 10 different strictness levels (Intersection over Union). For a Faster R-CNN model, this is a solid, respectable score for complex real-world data.



(a) Training Loss Curve



(b) Validation mAP Curve

Figure 5.14: Training loss curve and Validation mAP Curves

Classification Report for YOLOv11

In the Table 5.7 shows that the model Faster R-CNN achieved an overall accuracy of 62.1%, Precision 46.4%, Recall 46.6%, mAP_{50} 62.1%, mAP_{50-95} 35.5%. The F1 score of the model was 40.3%

Table 5.7: Evaluation Report of Faster R-CNN

Parameter	**Value**
Precision	46.4%
Recall	46.6%
mAP_{50}	62.1%
mAP_{50-95}	35.5%
F1-score	40.3
Object size AP (Small)	13.4%
Object size AP (Medium)	27.1%
Object size AP (Large)	46.4.3%

5.3 Comparative Analysis

in this thesis, three Convolutional Neural Network (CNN) models were implemented and evaluated based on their training and validation performance. Each model was trained for 70 epochs, and their performance was assessed using standard evaluation metrics shown in the Table 5.8. Among the models, the proposed approach using the YOLOv8 model achieved the

highest accuracy of 97.7%, demonstrating superior performance compared to the other models. The comparative analysis of testing and Validation accuracy across all models is presented in Figure 5.15.

5.3.1 Models Comparison and Analysis

To provide a more detailed understanding of model performance, Precision, Recall, F1 score, mAP_{50}, mAP_{50-95} and Accuracy score were analyzed for each models. The Table 5.8 and Figure 5.15 shows the Precision, Recall F1-Score, mAP50, mAP50-95 and Accuracy of each model.

Table 5.8: Evaluation Comparison between all Models

Model Name	**Precision**	**Recall**	**F1-Score**	mAP_{50}	mAP_{50-95}	**Accuracy**
YOLOv8	93.7%	93.9%	93.7%	97.5%	82.2%	97.7%
YOLOv11	93.1%	92.9%	93.0%	97.3%	81.1%	97.3%
YOLOv8	46.4%	46.6%	40.3%	62.1%	35.3%	62.1%

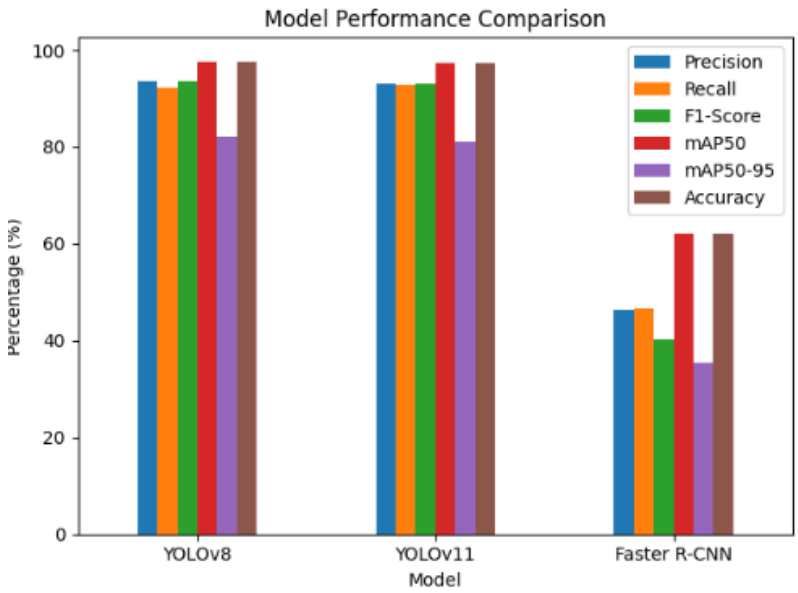


Figure 5.15: Evaluation Comparison between all Models

5.4 Summary

In this study, a total of three models were applied to the data set to evaluate their classification performance. Among all the models, the YOLOv8 architecture achieved the highest accuracy, reaching at 97.7%, thereby demonstrating superior effectiveness IN Road pothole detection task. Precision measures how accurately a model identifies positive sample without false positive. YOLOv8 Achieved a score of 93.7% for Precision for the ”road_pothole” class. Both YOLOv8 And YOLOv11 performed Very well for the detection.

CHAPTER 6
Conclusions and Future Works

6.1 Conclusions

This research successfully demonstrates the application of transfer learning with pre-trained Convolutional Neural Network (CNN) models for automated road pothole detection and classification. Among the three models evaluated, YOLOv8 achieved the best overall performance with an accuracy of 97%, Precision of 93.7%, Recall of 93.9%, mAP_{50} of 97.5%, F1-Score of 93.7%, and mAP_{50-95} of 82.2%. The study confirms that transfer learning significantly reduces training time and computational resource consumption while maintaining high detection accuracy. This makes it a highly viable approach for developing countries like Bangladesh, where the proposed low-cost system can aid authorities in timely maintenance prioritization and enhance driver safety through automated detection.

6.2 Limitations

Despite the high performance of the developed models, several limitations were observed during the course of this study:

- **Environmental Variations:** The model was primarily trained on datasets with clear visibility. Its performance may fluctuate under extreme weather conditions such as heavy monsoon rain, dense fog, or low-light night-time scenarios common in rural areas.
- **Static Perspective:** The current analysis is based on static images, which do not account for the motion blur and camera vibrations inherent in high-speed vehicle environments.

6.3 Future Recommendations

To enhance the utility and scalability of this research, the following directions are recommended for future work:

- **Real-Time Video Processing:** Future iterations should focus on deploying the model for real-time video stream analysis. Implementing object tracking algorithms (e.g., ByteTrack or DeepSORT) would ensure that a single pothole is not counted multiple times as the vehicle passes.
- **Integration with Edge Hardware:** Testing the model on dedicated edge computing hardware, such as NVIDIA Jetson Nano or Raspberry Pi with an AI accelerator, to create a standalone, plug-and-play dashboard device.
- **GPS-Linked Reporting System:** Developing a mobile application that integrates the detection model with GPS coordinates. This would allow for the automated creation of a ”Pothole Map” enabling municipal authorities to visualize damage clusters in real-time.
- **Depth Estimation:** Incorporating stereo-vision cameras or depth sensors to estimate the severity (depth and volume) of potholes, allowing for more precise maintenance prioritization.
- **Expansion of Road Anomalies:** Extending the dataset to include other types of road distress, such as alligator cracking, longitudinal cracks, and shoulder drop-offs, to provide a comprehensive road health monitoring system.

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APPENDIX

.1 Environment Setup and Library Verification

This appendix presents the Python code used to verify the GPU environment and install the required deep learning framework for pothole detection experiments.

```python
import os
import glob
from IPython.display import Image, display

# Check GPU availability
!nvidia-smi

# Get current working directory
HOME = os.getcwd()
print(HOME)

# Install Ultralytics YOLO framework
!pip install ultralytics

# Verify installation
import ultralytics
ultralytics.checks()
```

Listing 1: GPU Verification and Ultralytics Installation

.2 Dataset Preparation Using Roboflow

This appendix presents the Python code used to download and prepare the pothole detection dataset using the Roboflow platform. Roboflow was utilized for dataset versioning, annotation management, and export in YOLOv8-compatible format.

```python
# Install Roboflow library
!pip install roboflow
# Import Roboflow
from roboflow import Roboflow
# Initialize Roboflow with API key
rf = Roboflow(api_key="jaI2fhO2HO2aItbUNpPF")
# Access workspace and project
project = rf.workspace("mehedi-hasan-j8mol").project("
    road_potholes_project-g1a43")
# Select dataset version
version = project.version(7)
# Download dataset in YOLOv8 format
dataset = version.download("yolov8")
```

Listing 2: Roboflow Dataset Download for YOLOv8

.3 YOLOv8 Dataset Configuration File Verification

This appendix presents the Python code used to open and inspect the dataset configuration file (`data.yaml`). The file contains essential information such as dataset paths, class names, and the number of classes required for training the YOLOv8 model.

```
# Open and read the dataset YAML configuration file
with open('/kaggle/working/datasets/Road_potholes_project-7/data.yaml', 'r') as f:
    data = f.read()

# Display the original content of the file
print(data)
```

Listing 3: Reading YOLOv8 Dataset Configuration File

.4 Modification of YOLOv8 Dataset Configuration File

This appendix presents the Python code used to modify the `data.yaml` configuration file for the YOLOv8 training process. The file was updated to define the class name, number of classes, dataset paths, and Roboflow metadata to ensure compatibility with the training framework.

```
# Define updated dataset configuration
new_data = """
names:
- road_pothole
nc: 1
roboflow:
  license: CC BY 4.0
  project: road_potholes_project-g1a43
  url: https://universe.roboflow.com/mehedi-hasan-j8mol/road_potholes_project-g1a43/dataset/5
  version: 7
  workspace: mehedi-hasan-j8mol
train: /kaggle/working/datasets/Road_potholes_project-7/train
val: /kaggle/working/datasets/Road_potholes_project-7/train
test: /kaggle/working/datasets/Road_potholes_project-7/test
"""

# Write updated configuration to data.yaml
with open('/kaggle/working/datasets/Road_potholes_project-7/data.yaml', 'w') as f:
    f.write(new_data)

# Display the updated configuration file
!cat /kaggle/working/datasets/Road_potholes_project-7/data.yaml
```

Listing 4: Updating and Verifying YOLOv8 Dataset Configuration File

.5 Environment Setup Commands

```
!pip install fastapi kaleido python_multipart uvicorn
%cd {HOME}
%cd {dataset.location}
!pip install numpy==1.26.4 --force-reinstall
!pip install matplotlib==3.8.4 --force-reinstall
!pip install ultralytics --upgrade --no-cache-dir
```

Listing 5: Python and Environment Setup Commands

.6 YOLOv8 Training Commands

```
# Change to home directory
%cd {HOME}

# Train YOLOv8 model
!yolo task=detect mode=train model=yolov8m.pt \
    data={dataset.location}/data.yaml \
    epochs=70 imgsz=640 \
    name=yolov8_pothole_segment
```

Listing 6: Training YOLOv8 Model for Road Pothole Detection

.7 Visualizing Training Results

```
# Change to home directory
%cd {HOME}

# Display the confusion matrix image from YOLOv8 training results
from IPython.display import Image
Image(filename = f'{HOME}/runs/detect/yolov8_pothole_segment/
    confusion_matrix.png', width = 900)

# Display the loss curve of training results
Image(filename = f'{HOME}/runs/detect/yolov8_pothole_segment/results.
    png', width = 600)
```

Listing 7: Display YOLOv8 Confusion Matrix and loss curve

.8 YOlOv8 Validation Commands

```
!yolo task=detect mode=val model={HOME}/runs/detect/
    yolov8_pothole_segment/weights/best.pt data={dataset.location}/
    data.yaml
```

Listing 8: Display Validate YOLOv8 Model for Road Pothole Detection

.9 Visualization Validation Results

```
Image(filename = f'{HOME}/runs/detect/val2/confusion_matrix.png',
    width = 900)

Image(filename = f'{HOME}/runs/detect/val2/BoxF1_curve.png', width =
    600)

Image(filename = f'{HOME}/runs/detect/val2/BoxPR_curve.png', width =
    600)

Image(filename = f'{HOME}/runs/detect/val2/BoxP_curve.png', width =
    600)

Image(filename = f'{HOME}/runs/detect/val2/BoxR_curve.png', width =
    600)
```

Listing 9: Display Confusion Matrx, P, R, PR and F1-Score Confidence Curve

.10 YOlOv8 Test Commands

```
%cd {HOME}
!yolo task=detect mode=predict model={HOME}/runs/detect/
    yolov8_pothole_segment/weights/best.pt conf=0.25 source=/kaggle/
    working/datasets/Road_potholes_project-7/test/images
```

Listing 10: Display Test YOLOv8 Model for Road Pothole Detection

.11 YOlOv8 Test Image Result

```
Image('/kaggle/working/runs/detect/predict4/20240621_210411_jpg.rf.5
    ff2876d11ba7e3211bd819f42e8059d.jpg')
```

Listing 11: Display Test Image Result YOLOv8 Model for Road Pothole Detection

BIOGRAPHY

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Mehedi Hasan

Mehedi Hasan, born in Sarishabari, son of Md. Shafiqul Islam and Argina Akter is a National IT challenge,2020 1st Runner-up, Global IT Challenge,2021 2nd Runner-up and a dedicated computer enthusiast. He is pursuing his B.Sc. in Computer Science and Engineering from Bangladesh Army University of Science and Technology, Saidpur. He completed his SSC from Sarishabari Riaz Uddin Talukdar High School, Sarishabari, with a GPA of 4.22, and his HSC from Sarishabari Honours College, Sarishabari with a GPA of 4.50. In academics, Mehedi Hasan has demonstrated his commitment to advancing knowledge through his research endeavors. He is actively engaged in the research field of machine learning and deep learning. He has experience in C, and C++ and C#. He has a strong interest in Game Development and 3D Artist. He has also has experience in Unreal Engine, Unity and Blender. Mehedi Hasan played an important role in his thesis research, Road Pothole Detection Using Machine Learning.

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She has experience in Java, C, and C++. She has a strong interest in web design. She also has experience in Python, PyTorch, OpenCV and AI model optimization. She actively participates in research projects, workshops and coding competitions, aiming to advance intelligent automation systems.

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